THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(d)** 3 x 10^5	2 x 10^30 / 6 x 10^24.
15	**(b)** 50 per cent	30/60.
16	**(c)** 2 : 1	75/25 = 3; 60/40 = 1.5; ratio 2 : 1.
17	**(d)** S^{2-}	Z = 16 is sulphur, 2 extra electrons.
18	**(c)** u / sqrt(2)	v squared = u squared - 2g(h/2) with h = u squared/2g.
19	**(c)** 1 : 4	a is inversely proportional to m.
20	**(c)** 100 J	Half converted to PE.
21	**(d)** 0.68 m and 3 m	lambda = v/f.
22	**(c)** hypotonic to the cell	Water entered.
23	**(a)** accumulation of food	Phloem blocked.
24	**(d)** Fungi	Hyphae and chitin.
25	**(d)** 14 March	Mid cycle.
26	**(c)** 4000 J	400 W x 10 s.
27	**(d)** the theory is strongly supported but remains open to tests	Science is provisional.
28	**(a)** only one component sublimes	Different behaviour on heating.
29	**(a)** 4.4 g	Mass of gas escaped.
30	**(b)** 24.3 u	(1896 + 250 + 286)/100 = 24.32.
31	**(c)** 8 km/h	24 km/ 3 h.

32	**(a)** the pull of the book on the Earth	Third-law pair.
33	**(b)** 250 J	mgh.
34	**(d)** 24 m/s	0.6 x 40.
35	**(a)** synthesis of ATP	Mitochondria do ATP.
36	**(b)** glandular epithelium	Secretion.
37	**(b)** They are in the same class and phylum	Higher ranks shared.
38	**(a)** the one with an eye grows	Bud needed.
39	**(c)** algal bloom and fish kill	Eutrophication.
40	**(d)** A is false but R is true.	A is false (theories can be refined); R is true.
41	**(a)** Both A and R are true and R is the correct explanation of A.	Some readings are slightly too high and some too low, so the mean reduces the effect of random errors.
42	**(a)** Both A and R are true and R is the correct explanation of A.	Sand stays on the filter paper while the salt solution passes through; the difference in solubility explains the method.
43	**(a)** Both A and R are true and R is the correct explanation of A.	Since the number and kind of atoms stay the same, the total mass is conserved.
44	**(a)** Both A and R are true and R is the correct explanation of A.	Alpha particles pass through the empty space and are deflected only when they come close to the tiny nucleus.
45	**(a)** Both A and R are true and R is the correct explanation of A.	If a body returns to its starting point, the shortest distance between start and finish is zero, although it has travelled a path.
46	**(a)** Both A and R are true and R is the correct explanation of A.	This is inertia of motion, which is the reason for the forward jerk.
47	**(a)** Both A and R are true and R is the correct explanation of A.	There is no displacement, so $W = Fs = 0$.
48	**(a)** Both A and R are true and R is the correct explanation of A.	With no particles to vibrate, no compressions and rarefactions can form.
49	**(a)** Both A and R are true and R is the correct explanation of A.	ATP is the energy currency of the cell, which is why mitochondria are called powerhouses.
50	**(a)** Both A and R are true and R is the correct explanation of A.	Growth requires new cells, which are produced by dividing meristematic cells.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** wings or hairs	Light structures.
2	**(b)** the ground is heated by sunlight and warms the air	Surface heating.
3	**(d)** 5.4 cm	Subtract the zero error.
4	**(a)** 32 g	Water 100 g holds 32 g; crystals 64 - 32.
5	**(b)** 3 : 8, CO_2	Atoms ratio 2.4/12 : 6.4/16 = 1 : 2.
6	**(d)** 20 and 16	X lost 2, Y gained 2.
7	**(b)** 1.5 t	v = at; retard time = at/2a = t/2.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** lignin	Lignified walls.
13	**(a)** dolphins have lungs, hair and mammary glands	Mammalia vs Pisces.
14	**(b)** oviduct	Tube.
15	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
16	**(d)** dosage of a medicine for a child	Safety critical.
17	**(b)** iron and sulphur	Only iron is magnetic.
18	**(d)** 78.5	52.5/47.5 x 71 = 78.5.
19	**(d)** 20	10x + 11(100 - x) = 1080.
20	**(a)** 4.8 km/h	Harmonic mean.
21	**(c)** 20 m/s	a = 4 m/s squared; v = 20 m/s.
22	**(c)** 3h/4	KE = 1/4 of mgh.
23	**(c)** 2 s	680/340.
24	**(a)** hypertonic to A and hypotonic to B	Different internal concentrations.
25	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
26	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
27	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
28	**(b)** increase	More absorption.
29	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
30	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
31	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.

32	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
33	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
34	**(b)** reducing the friction by rollers	Reduce friction.
35	**(a)** 4 m/s	sqrt(2 x 10 x 0.8).
36	**(c)** 86 m	vt.
37	**(b)** bacterial	Prokaryote.
38	**(b)** xylem is intact	Water still moves.
39	**(b)** Gymnosperms	Cones with naked seeds.
40	**(d)** 13 to 15 March	Around ovulation.
41	**(a)** Both A and R are true and R is the correct explanation of A.	Interbreeding with fertile offspring is the defining feature of a species.
42	**(a)** Both A and R are true and R is the correct explanation of A.	With one parent and no gamete fusion, the offspring are clones of the parent.
43	**(a)** Both A and R are true and R is the correct explanation of A.	Albedo is the fraction of sunlight reflected; snow reflects a large fraction, so its albedo is high.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the SI unit does not explain why the least count is 1 mm (it is the smallest division on the scale).
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements, but R does not explain why a fractionating column is needed in A.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the fixed ratio is explained by the law of constant proportions, not just by water being a compound.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the discovery of the neutron is not the reason for the model's failure (a revolving electron should lose energy and spiral into the nucleus).
48	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements about graphs, but the second does not explain the first.
49	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but they are unrelated statements.
50	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the definition of power does not explain the unit of energy.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** a single nucleus and spindle shape	Unstriated.
2	**(c)** belong to the same genus but different species	Same genus Canis.
3	**(a)** cambia of stock and scion are in contact	Tissue contact.
4	**(a)** absorption of UV by ozone	Ozone heating.
5	**(a)** check the experiment and the assumptions in the model	Investigate.
6	**(c)** dissolve in water, filter, evaporate	Sugar dissolves.
7	**(b)** 27.3 per cent	12/44 x 100 = 27.3.
8	**(a)** electron	An electron is about 1/1836 of the mass of a proton.
9	**(a)** 24 m/s	2 x 20 x 30/50.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(c)** A is true but R is false.	A is true; R is false: KE is zero there.
12	**(d)** A is false but R is true.	A is false (loudness does not depend on frequency); R is true.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(c)** dendrite to axon	Direction.
15	**(b)** Plantae	Plant cell.
16	**(b)** hooks or fleshy fruit	Burrs.
17	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
18	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
19	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
20	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
21	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
22	**(d)** A is false but R is true.	A is false: acceleration is g throughout; R is true.
23	**(d)** 10 m/s squared downward	Only gravity acts.
24	**(b)** 4 : 1	KE = p squared/2m.
25	**(c)** 1 : 3	Distance proportional to time.
26	**(a)** 10	Mitosis keeps 20; meiosis halves.
27	**(c)** safranin	Red lignin.
28	**(d)** have hair and mammary glands	Mammals.
29	**(d)** prostate and seminal vesicles	Semen.
30	**(c)** 100	2200/22.
31	**(a)** 15 per cent	3 x 5 per cent.
32	**(a)** the one with R_f 0.9	Largest R_f.
33	**(c)** 310 u	120 + 62 + 128.
34	**(d)** 5	Z = 15: 2, 8, 5.
35	**(d)** 1 : 3 : 5	Odd numbers.

36	**(a)** the same for both	F = change in p/t.
37	**(a)** 72 J	F = m(g + a) = 24 N.
38	**(b)** 20 m	t1 = sqrt(2h/g), t2 = h/v: h = 20 gives 2 + 0.06.
39	**(a)** energy release	Respiration.
40	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
41	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements but R does not explain A.
42	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the composition of the wall does not explain why animal cells lack it.
43	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the nature of xylem does not explain the function of phloem.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; R states the rule of writing but does not explain why the human name is *Homo sapiens*.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true but describe different organisms and methods.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but the composition of the atmosphere does not explain why CO_2 is a greenhouse gas.
47	**(c)** A is true but R is false.	Models keep only the features that matter for the question, so they are not exact copies; R is false.
48	**(c)** A is true but R is false.	Suspension particles are large and settle on standing, so R is false.
49	**(c)** A is true but R is false.	Aluminium has valency 3 and oxygen has valency 2, so criss-crossing gives Al_2O_3; R is false.
50	**(c)** A is true but R is false.	A is true; electrons are about 1/1836 of the mass of a proton, so R is false.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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